

Dear Editors of Evolutionary Human Sciences,

I am grateful for your letter of 27 July 2026 regarding manuscript EHS-2026-0079, suggesting that the appropriate format for our submission is a Commentary on Itao (2026). I have now revised the work accordingly.

The enclosed Commentary, “Two pathways, many clocks: a commentary on Itao (2026),” is approximately 1,000 words and contains one composite figure (Figure 1), within the Commentary limits specified in the Author Instructions.

This Commentary engages directly with Itao (2026, “Two universal pathways in demographic transition,” *Evolutionary Human Sciences*, <https://doi.org/10.1017/ehs.2026.10054>), whose statistical-physics reading of demography—identifying two conserved quantities that organise the crude birth rate against life expectancy—is an imaginative cross-disciplinary contribution. My aim is not to overturn that framework but to build on it: to ask how far the reported regularity extends once national transitions are aligned on their own timing and the analysis is pushed to sub-national scales.

Using the same public data, I show three complementary results. (1) National fertility transitions are highly asynchronous (onsets spanning 1828–2013), so a single mid-century change point is best read as a summary of aggregated timing rather than a synchronised global event. (2) Re-aligning each country on its own transition onset sharpens the first conserved quantity considerably, while the second quantity does not tighten—so the two ‘pathways’ are asymmetric. (3) Below the national level, regions of the same country at nearly identical life expectancy differ in birth rate by as much as the cross-national spread the law treats as universal. These findings point to event-time alignment and hierarchical modelling as natural complements to the original programme.

All numbers and figures regenerate from public data and public code with a single command; the repository is at <https://github.com/bougtoir/demographic-transition-universality> and will be archived on acceptance. No value in the manuscript is hand-entered.

This Commentary is original, has not been published previously, and is not under consideration elsewhere. I have no conflicts of interest to declare and the work received no specific funding. I am the sole author.

Thank you for considering this revised submission. I look forward to your response.

Sincerely,

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